01 / THE CORE IDEA

Ohm's Law

A small equation. A useful way to see an electrical circuit.

The relationship

For an ohmic conductor, current is proportional to the potential difference across it when temperature and other physical conditions remain constant.

$$V = I \times R$$

Potential difference = current x resistance

Know your quantities

V : potential difference, measured in volts (V)

I : electric current, measured in amperes (A)

R : resistance, measured in ohms (symbol: Greek omega)

The graph tells a story

For a V-versus-I graph, slope = resistance R.

For an I-versus-V graph, slope = 1/R.

Check the axes before interpreting a slope.

02 / MAKE THE CONNECTION

From idea to example

Start with what you know. Let the units guide you.

Worked example

A 6 ohm resistor is connected across a 12 V supply.

Find the current, assuming the resistance stays constant.

$$I = V / R = 12 / 6 = 2 \text{ A}$$

The current through the resistor is 2 amperes.

What changes if the voltage doubles?

At the same resistance, I = 24 / 6 = 4 A.

Doubling voltage doubles current only if resistance is constant.

A common misconception

Ohm's law does not describe every electrical device.

A diode is non-ohmic. A filament lamp heats up, changing its resistance. Do not assume constant R as temperature changes.

Extension: 2 ohms + 4 ohms in series = 6 ohms total.

03 / PAUSE AND PRACTICE

Make it your own

A short checklist for careful problem solving.

Before calculating

1. Identify the known values and the quantity you need.
2. Convert units: 500 mA = 0.5 A.
3. Choose V = IR, I = V/R, or R = V/I.
4. Check the units and the physical conditions.

V = 0.5 x 10 = 5 V

A 500 mA current through 10 ohms needs a potential difference of 5 V.

A quick check

A straight V-I line through the origin indicates constant resistance over the measured range. A conductor is ohmic only under conditions where this proportionality holds.

Explain it to yourself

Why does a greater resistance mean less current at the same voltage? Use I = V/R to connect the words and the equation.